

Ashley Sample Management Agent - Product Requirements Document (PRD)

Document Info

- **Product Name:** Ashley Sample Management Agent (Pre-PLM AI Agent)
- **Version:** 2.0.0
- **Last Updated:** 2026-05-19

1. Product Overview

1.1 Background & Goals

The R&D sampling process involves extensive product information entry, modification, querying, and report generation. Traditional methods rely on manual spreadsheets or complex ERP systems, presenting the following pain points:

Pain Point	Description
Scattered Information	Product info scattered across WeChat groups, emails, on-site photos, notes, etc.
Inefficient Entry	Traditional methods are complex with usage limitations
Delayed Response	Checking sampling progress requires waiting for manual report compilation and PPT editing
Disconnected from AI	AI Agents have become standard for automation; current xwork approach is outdated

Product Goal: Build a **low-barrier, real-time** sample management assistant based on Feishu Bot + AI Agent, enabling R&D, procurement, and management to manage product information through natural conversation.

1.2 Target Users

Role	Use Case	Core Need
R&D/Procurement	Quick product entry at sample sites, photo upload, quick modifications, queries, auto PPT generation	Fast response, automated generation
Management	Track sample status, generate report PPTs	One-click reports, real-time progress
Suppliers	Receive rectification requirements, view issue feedback	Clear problem images and requirements

1.3 Product Positioning

Ashley Sample Management Agent = Feishu Bot + LLM Intent Understanding + Function Calling + Structured Data Storage + Auto Report Generation

2. Functional Requirements

2.1 Core Modules

2.1.1 Product Entry & Modification

Description: Auto-extract and save product attributes to database through chat conversation.

User Stories:

- User can say "Longwei sofa, FOB 385, cube 17.9" and system auto-recognizes and saves

Supported Fields:

Category	Fields	Description
Basic Info	series_no, product_name, category, vendor, market	Product identity & attribution
Commercial	fob, fca, ddp, first_sale, ws_price, rt_price	Pricing

Logistics	product_size, packing_size, cube, gross_weight, loading_qty	Dimensions & loading
Changes	product_changes, decision	Product changes & decisions

Interaction Flow:

User Input → AI Intent Parsing → Call entry/modify tool to generate confirmation card → User confirms → Write to database

Confirmation Card Example:



About to save the following:
Series No: UE0890
Product Name: Nordic Sofa
FOB: \$385
Cube: 17.9 CBM

Reply "Confirm" to save, or "Cancel" to abort.

2.1.2 Multimodal Information Extraction

Description: Auto-identify uploaded image types, extract product attributes and issue descriptions.

Image Type Classification:

Type	Identifier	Description	Processing
Product	Product	Actual sample image	Extract visible attributes
Rendering	Rendering	3D rendering	Associate with product series
Material	Material	Material sample	Extract material info
Note	Note	Image with text annotations	OCR text extraction
Problem	Problem	Issue-marked image	Extract problem descriptions

Interaction Flow:

User sends image → AI identifies type → Extract info/issues → Associate with product → Store to cloud

2.1.3 Product Query & Search

Description: Support single product detail queries and range searches.

Single Product Query:

- Input: "How is UE0890 doing"
- Output: Display all recorded info including prices, dimensions, change records, associated images

Range Search:

- Input: "What products does Longwei have"
- Output: List all product series under that supplier

Supported Search Dimensions:

- By product number / name
- By supplier
- By time range (last N days)
- By market (US/EU/Asia)

2.1.4 Auto PPT Report Generation

Description: One-click generation of multiple product report PPT versions.

Report Versions:

Version	ID	Use Case	Focus
Product Application	go_apply	New product approval	Complete product info, pricing, images
Trip Report	trip_note	Business trip review	Product images, changes, issues
Factory Rectification	factory	For suppliers	Problem images, requirements (CN/EN)

All	all	Complete archive	All information
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Filter Criteria:

- Batch generation: By supplier, market, date

Output Format:

- PPTX file (auto-layout, smart multi-image arrangement)
- Sent to user via Feishu file card

2.1.5 Image View & Management

Description: View product-associated images of various types, support deletion.

Interaction Flow:

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User: "Show UE0890 problem images"  
Bot: Display image list card (index + thumbnail + description)  
User: "Delete 2"  
Bot: Confirm deletion of image #2
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2.2 Auxiliary Features

2.2.1 Context Memory

- Auto-record current conversation focus (product being discussed)
- Support pronoun resolution: "What's its FOB" → auto-associate with previous product

2.2.2 No-Code Product Handling

- Users can enter without providing a number; system auto-generates placeholder (NOCODE-xxx)
- Later can add formal number via "Longwei sofa's number is UE0890"

2.2.3 User Permission Management

Role	Permission Scope
internal	Internal staff, can view/enter all products
supplier	Supplier, can only view related products

3 Non-Functional Requirements

5. Non-Functional Requirements

3.1 Performance Requirements

Metric	Requirement
Text message response	< 15 seconds
Image processing response	< 20 seconds
PPT generation (single)	< 60 seconds
PPT generation (batch 10)	< 180 seconds

3.2 Availability Requirements

- 7×24 service
- Core functions (entry/query) can fall back to rule engine when LLM interface fails

3.3 Security Requirements

- Prevent system Prompt leakage
- Prevent cross-user data access
- Delete operations must go through confirmation flow
- Optimistic locking to prevent concurrent modification conflicts

4. Data Model

4.1 Core Entities

ProductSeries
├─ series_no (PK)
├─ product_name
├─ vendor
├─ Commercial info (fob, ddp, cube...)
└─ Change records (product_changes)
Products (SKU)
├─ sku_no
├─ series_no (FK)
└─ Item attributes

ProductImages
└─ target_type (series/product)
└─ image_type (Product/Note/Material/Rendering/Problem)
└─ cloud_url
ProductUpdates (Change History)
└─ field_name
└─ old_value → new_value
└─ operator/timestamp

4.2 Data Flow

Feishu Message → Deduplication → AI Parsing → Confirmation Card → Write to xwork API → Cloud Storage
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5. Interface Dependencies

External System	Purpose	Interface Type
Feishu Open Platform	Message send/receive, file upload	WebSocket + REST API
OpenAI / Azure OpenAI	LLM intent understanding, extraction	REST API
xwork Platform	Data storage, attachment management	REST API
Azure Blob Storage	Image cloud storage	Via xwork wrapper

6. Release Plan

6.1 V1.0 (Entry, Query - Released)

- ✓ Product entry & modification
- ✓ Multimodal image processing
- ✓ Single product query & range search

6.2 V2.0 (Current Version)

- ✓ xwork integration

- ✓ Fuzzy entry
- ✓ Batch image write
- ✓ Image association
- ✓ Image viewing
- ✓ PPT generation

6.3 V2.1 (Planned)

- ☐ Editable HTML functionality
 - ☐ Image selection feature
 - ☐ Data write to xwork PLM system
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